Day 13 Activity: Reshaping data

Thursday, June 4, 2026

Your Name Here

Setup

```r
library(tidyverse)
library(palmerpenguins)
```

Today's verbs are `pivot_longer()` and `pivot_wider()`. We'll use a few small built-in tables plus `penguins`.

Part 1

Look at `table1` and `table2` (both built into tidyr):

```r
table1
```

```
# A tibble: 6 x 4
  country      year  cases population
  <chr>       <dbl>  <dbl>      <dbl>
1 Afghanistan  1999    745   19987071
2 Afghanistan  2000   2666   20595360
3 Brazil       1999  37737  172006362
4 Brazil       2000  80488  174504898
5 China        1999 212258 1272915272
6 China        2000 213766 1280428583
```

```r
table2
```

```
# A tibble: 12 x 4
   country      year type            count
   <chr>       <dbl> <chr>           <dbl>
 1 Afghanistan  1999 cases             745
 2 Afghanistan  1999 population   19987071
 3 Afghanistan  2000 cases            2666
 4 Afghanistan  2000 population   20595360
 5 Brazil       1999 cases           37737
 6 Brazil       1999 population  172006362
 7 Brazil       2000 cases           80488
 8 Brazil       2000 population  174504898
 9 China        1999 cases          212258
10 China        1999 population 1272915272
11 China        2000 cases          213766
12 China        2000 population 1280428583
```

1a. For each, describe what **one row** represents.

Table 1: [Answer here.]

Table 2: [Answer here.]

1b. Which one is **tidy** (each variable a column, each observation a row)? Which is the **narrower** of the two, and how can you tell?

[Answer here.]

Part 2

Build this wide table by hand:

```r
bp_wide <- tribble(
  ~subject, ~before, ~after,
  "BHO", 160, 160,
  "GWB", 115, 135,
  "WJC", 105, 145
)
bp_wide
```

```
# A tibble: 3 x 3
  subject before after
  <chr>    <dbl> <dbl>
1 BHO        160   160
2 GWB        115   135
3 WJC        105   145
```

2a. Predict first. If you pivot `before` and `after` into a long form, how many **rows** will the result have? How many **columns**?

Prediction:

> [Pred. # Rows here]
>
> [Pred. # Cols here]

2b. Now pivot `bp_wide` to long form. Use `names_to = "when"` and `values_to = "sbp"`. Check against your prediction.

```
# Your code:
```

Part 3

The `billboard` data (in tidyr) has one column per chart week (`wk1`, `wk2`, …).

3a. Pivot all the `wk` columns into `week` / `rank`, dropping the empty weeks. (Hint: `cols = starts_with("wk")`, `values_drop_na = TRUE`.)

```
# Your code:
```

3b. Add `mutate(week = parse_number(week))` so `"wk3"` becomes `3`, then plot `rank` (y) against `week` (x), one faint line per `track`. Use `scale_y_reverse()` so rank 1 sits on top.

```
# Your code:
```

One sentence: what does the plot show about how long songs stay on the charts?

> [Answer here.]

Part 4

Start from a long summary table:

```
penguin_summary <- penguins |>
  filter(!is.na(sex)) |>
  group_by(species, sex) |>
  summarise(avg_mass = mean(body_mass_g, na.rm = TRUE), .groups = "drop")
penguin_summary
```

```
# A tibble: 6 x 3
  species   sex    avg_mass
  <fct>     <fct>     <dbl>
1 Adelie    female    3369.
2 Adelie    male      4043.
3 Chinstrap female    3527.
4 Chinstrap male      3939.
5 Gentoo    female    4680.
6 Gentoo    male      5485.
```

4a. If you widen this with `names_from = sex`, `values_from = avg_mass`, how many rows and columns will you get?

Prediction:

> [Pred. # Rows here]
>
> [Pred. # Cols here]

4b. Pivot, then add a column `diff` = `male - female`.

```
# Your code:
```

Which species has the biggest mass gap between sexes?

> [Answer here.]

Part 5

5a. Take your widened penguin table from Part 4 (before the `diff` column) and pivot it **back** to long form. Do you recover the original `penguin_summary`? (`pivot_longer()` and `pivot_wider()` are inverses.)

```
# Your code:
```

5b. Someone tries to make `bp_wide` longer like this and gets a strange result:

```
bp_wide |>
  pivot_longer(
    cols = c(subject, before, after),
    names_to = "when",
    values_to = "sbp"
  )
```

Run it. Why is this "too narrow" / wrong? What single change to `cols` fixes it?

> [Answer here.]

Part 6 — Choose the shape

For each goal, say whether you'd want the data **wide** or **narrow**, and which pivot gets you there.

1. Plot each country's TB cases over time, one line per country.
2. Compute, for each country, the change in cases from 1999 to 2000 as a single new column.
3. Group by year and sum the cases across all countries.

> 1. [Answer here.]
> 2. [Answer here.]
> 3. [Answer here.]

Code appendix

```r
library(tidyverse)
library(palmerpenguins)
table1
table2
bp_wide <- tribble(
  ~subject, ~before, ~after,
  "BHO", 160, 160,
  "GWB", 115, 135,
  "WJC", 105, 145
)
bp_wide
# Your code:

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penguin_summary <- penguins |>
  filter(!is.na(sex)) |>
  group_by(species, sex) |>
  summarise(avg_mass = mean(body_mass_g, na.rm = TRUE), .groups = "drop")
penguin_summary
# Your code:

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bp_wide |>
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  )
```